

# 1 Solution — GIAM 5.3

## Problem 9

### 1.1 Problem

Give an inductive proof that  $\forall n \in \mathbb{N}, 6 \mid 2n^3 - 2n - 14$ .

### 1.2 Answer

False:  $2n^3 - 2n - 14 \equiv 4 \pmod{6}$  for every  $n \in \mathbb{N}$

The statement cannot be proved because it fails at every  $n$ . The smallest counterexample is  $n = 0$ , where the value is  $-14$ . The inductive step is perfectly valid; it is the basis that fails, and the step then carries the failure to every  $n$ . No value is divisible by  $6$ , checked for  $|n|$  up to  $5000$ .

### 1.3 The Disproof

To refute a universal statement one instance is enough. Write  $f(n) = 2n^3 - 2n - 14$ . At  $n = 0$ ,

$$f(0) = -14 = 6 \cdot (-3) + 4,$$

so  $6 \nmid f(0)$ , and the claim  $\forall n \in \mathbb{N}, 6 \mid f(n)$  is false. The same happens at  $n = 1$ , where  $f(1) = -14$  again, and at  $n = 2$ , where  $f(2) = -2$ .

Induction does prove the correct statement, which shows the failure is total.

**Theorem.** For every natural number  $n$ ,  $f(n) \equiv 4 \pmod{6}$ , so  $f(n) - 4$  is a multiple of  $6$ .

**Proof.** By induction on  $n$ .

**Basis.** At  $n = 0$ ,  $f(0) - 4 = -18 = 6 \cdot (-3)$ .

**Inductive step.** Let  $k$  be a particular but arbitrarily chosen natural number and suppose  $f(k) - 4 = 6m$  for some integer  $m$ . Then

$$\begin{aligned} f(k+1) &= 2(k^3 + 3k^2 + 3k + 1) - 2(k+1) - 14 \\ &= (2k^3 - 2k - 14) + 6k^2 + 6k \\ &= f(k) + 6k(k+1), \end{aligned}$$

so  $f(k+1) - 4 = 6m + 6k(k+1) = 6(m + k^2 + k)$ , and  $m + k^2 + k$  is an integer.

Basis and step together give the result for every  $n$ . ■

Since every value leaves remainder 4, none leaves remainder 0: the original statement is false for *every* natural number, not just for  $n = 0$ .

## 1.4 The Picture

The step links every value to the next, but the first link is never set off.

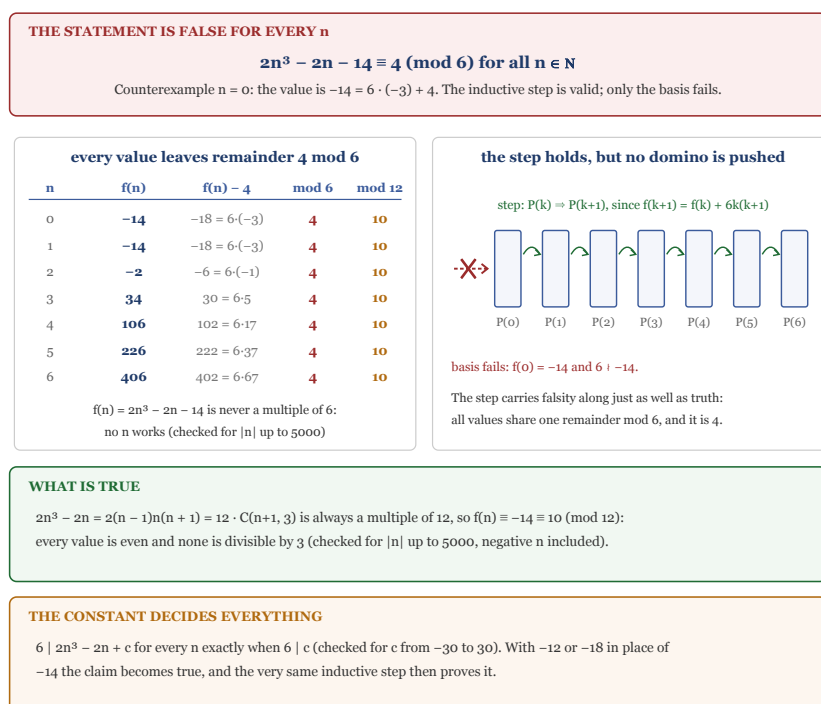


Figure 1.1: A diagram in four parts. A red band states that two  $n$  cubed minus two  $n$  minus fourteen is congruent to four mod six for every natural  $n$ , with counterexample  $n$  equal to zero. The left panel tabulates  $f$  of  $n$  for  $n$

from zero to six, each with remainder four mod six and ten mod twelve. The right panel draws dominoes  $P$  of zero to  $P$  of six joined by valid step arrows, with a crossed-out push before the first, since  $f$  of zero is minus fourteen. A green band states that  $f$  of  $n$  is congruent to ten mod twelve; an orange band notes that six divides two  $n$  cubed minus two  $n$  plus  $c$  for every  $n$  exactly when six divides  $c$ .

## 1.5 Where an Attempted Proof Goes Wrong

Someone attempting the requested proof would find that the inductive step works:

$$6 \mid f(k) \implies 6 \mid f(k) + 6k(k+1) = f(k+1).$$

That implication is genuinely valid for every  $k$ ; the algebra is correct. (Its premise also happens never to hold, so it is vacuously true as well.) But without a valid basis there is nothing for the step to act on. This is the lesson of 5.1 Problem 4 in its starkest form: there a missing basis left half the numbers unproved, while here the basis is false and so nothing is proved at all.

## 1.6 Remark — The Step Carries Falsity Too

The identity  $f(k+1) = f(k) + 6k(k+1)$  says more than “divisibility is passed along”. It says  $f(k+1)$  and  $f(k)$  always leave the *same* remainder on division by 6. So all the values share one remainder, and only the basis decides which. If one value were a multiple of 6, all of them would be; since  $f(0)$  is not, none is.

This is why checking the basis is not a formality. An inductive step on its own shows only that the property is *hereditary*: it is either true from the basis onward or, as here, never established at all. In this case the step works just as well with “ $6 \nmid f(k)$ ” as the hypothesis.

## 1.7 Remark — What Is True Modulo 12

Factoring the cubic part,

$$2n^3 - 2n = 2(n-1)n(n+1) = 12 \binom{n+1}{3},$$

checked for  $n$  up to **2000**. A product of three consecutive integers is divisible by **6** (as in 5.3 Problem 1's sharper remark), so  $2n^3 - 2n$  is a multiple of **12**, and

$$f(n) \equiv -14 \equiv 10 \pmod{12}.$$

So every value is even, and no value is divisible by **3**. That also holds for negative  $n$ , all checked for  $|n|$  up to **5000**. Indeed the step adds  $6k(k+1)$ , which is a multiple of **12** because  $k(k+1)$  is even.

## 1.8 Remark — The Constant Decides Everything

Since  $2n^3 - 2n$  is always a multiple of **6**,

$$6 \mid 2n^3 - 2n + c \text{ for every } n \iff 6 \mid c,$$

and a search over  $c$  from **-30** to **30** finds exactly the multiples of **6**. With **-12** or **-18** in place of **-14** the statement would be true, and the inductive step above would prove it unchanged, with only the basis checked differently. With **-14**, the basis is exactly where the attempt fails, and there is no way to repair it.